An empirical example: *Rhinoclemmys* data

Daniel R. Miranda-Esquivel

2024 / 07 / 20

To evaluate the effect of branch length on the area(s) chosen for conservation, I conduct a Phylogenetic Diversity (PD) analysis for the *Rhinoclemmys* genus. The topology I will use corresponds to a Total Evidence analysis from Romero-Alarcon (2020), and the distribution will be modified from Le and McCord (2008).

Read the data: the distribution, and the tree

```
## For reproductibility purposes

set.seed(121)


options(warn=-1)
library(ggplot2)
suppressMessages(library(ggtree))
#~ library(ape)
suppressMessages(library(blepd))

suppressMessages(library(phytools))

#~ library(ggtree)
##
## Just in case, if you want to install ggtree.
##
### https://bioconductor.org/packages/release/bioc/html/ggtree.html
##
####
#  if (!require("BiocManager", quietly = TRUE))
#      install.packages("BiocManager")
#  BiocManager::install("ggtree")

## Version

cat("Analyses made with blepd version:",unlist(packageVersion("blepd")))
```

```
## Analyses made with blepd version: 0 3 0 2024 7 16
```

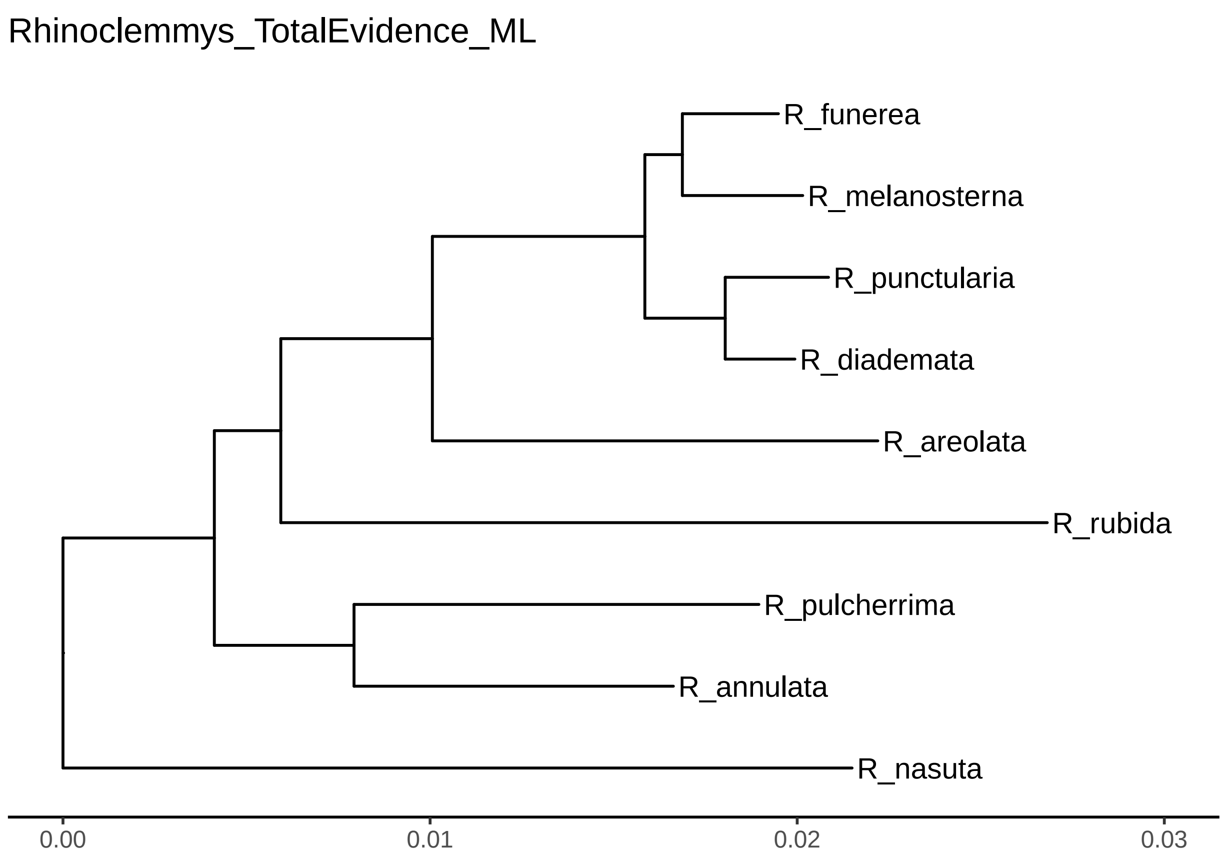
```
## Create an object to place the distribution and the tree

RhinoclemmysData <- list()

## Read data
```

```r
## distribution is a labeled csv file, areas by terminals

#setwd("./csv/")

## getwd()

csvFile <- list.files(pattern="csv")

## csvFile

### the functions use a matrix object for the distributions

RhinoclemmysData$distribution <-  as.matrix(read.table(csvFile,
                                  stringsAsFactors=FALSE,
                                  header=TRUE,
                                  row.names=1,
                                  sep=",")
                                  )


print(t(RhinoclemmysData$distribution))
```

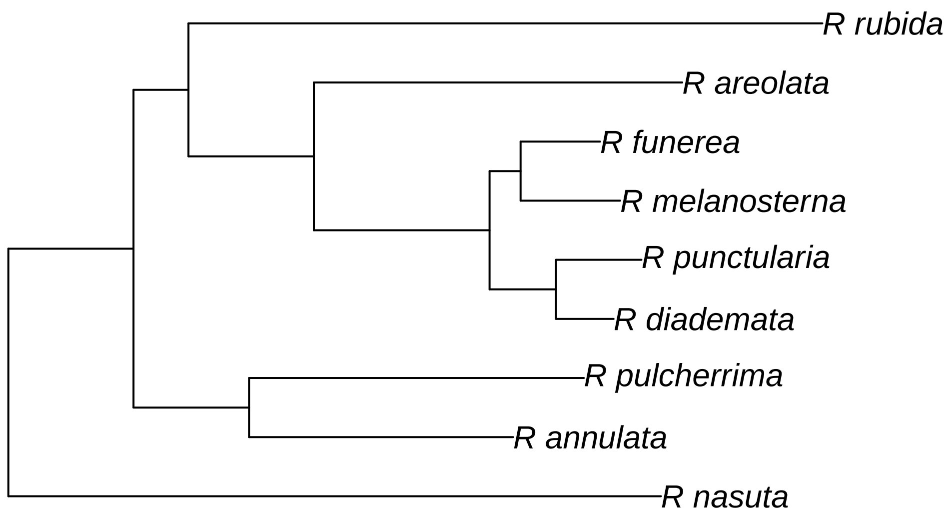
```
##                 Al An-As Cho Nsa Nuh Ph Pl
## R_annulata       1     0   1   0   0  1  0
## R_areolata       1     0   0   0   0  0  0
## R_diademata      0     0   0   1   0  0  0
## R_funerea        1     0   0   0   1  0  0
## R_melanosterna   0     0   1   0   0  1  0
## R_nasuta         0     0   1   0   0  0  0
## R_pulcherrima    0     0   0   0   1  0  1
## R_punctularia    0     1   0   0   0  0  0
## R_rubida         0     0   0   0   0  0  1
```

```r
## tree(s) in nexus or newick format

##setwd("../tree/")

treeFiles <- list.files(pattern=".tre")

##treeFiles

RhinoclemmysData$tree  <-    read.tree(treeFiles)


## name of tree(s)

treeFiles <- gsub(".tre","",treeFiles)



## Plotting

#par(mfrow=c(2,1))
```

```r
## The tree

## using ggtree

# RhinoclemmysData$tree <- reorder(RhinoclemmysData$tree, order = "cladewise")

plotTree <-  ggtree(RhinoclemmysData$tree, ladderize=TRUE,
                    color="black", size=0.51, linetype="solid") +
             geom_tiplab(size=4, color="black") +
             xlim(0,0.030) +
             theme_tree2() +
             ggtitle(treeFiles[1])


##

print(plotTree)
```

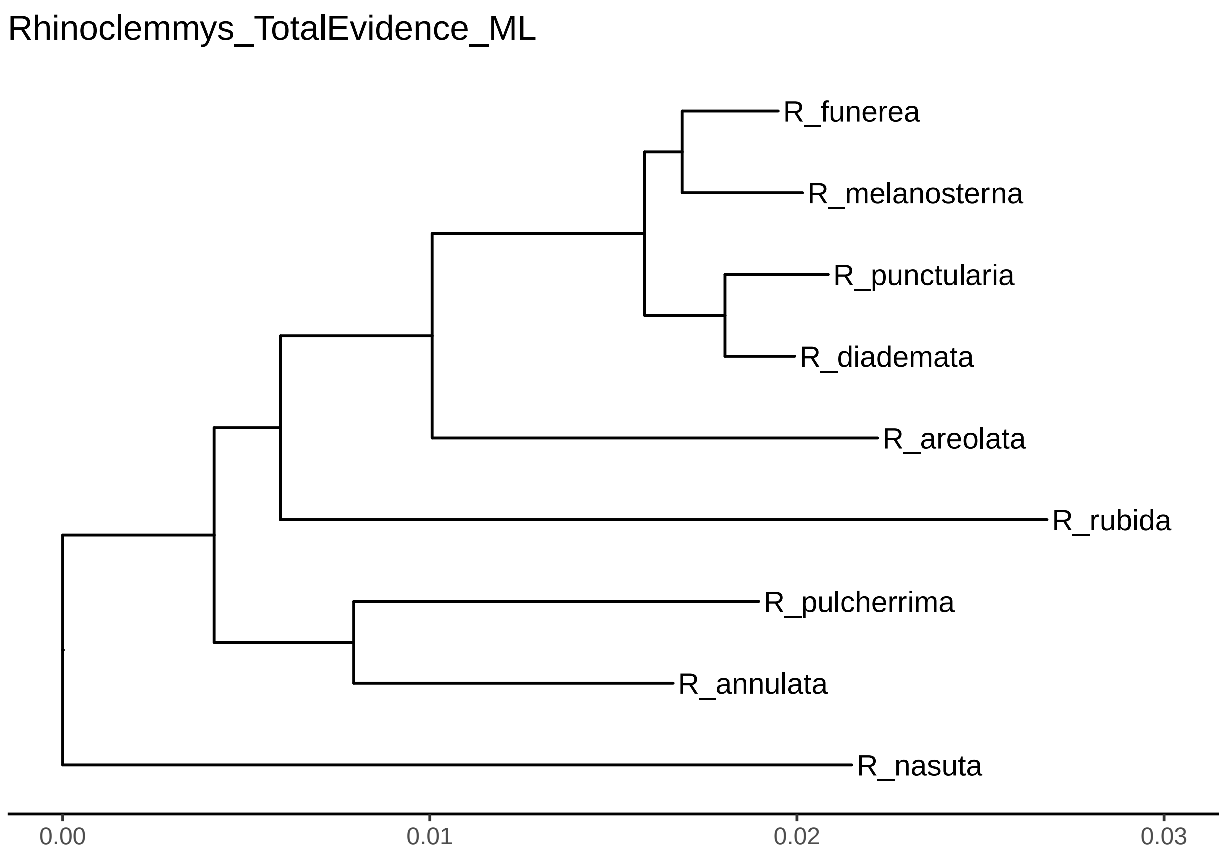



```r
#~ Alternatively, we can plot the trees using APE

##
##

plot.phylo(RhinoclemmysData$tree)
```




```
##

####nodelabels()
####tiplabels()


## to plot the distribution, we must transform it into a data.frame
## object

## distXY <- matrix2XY(RhinoclemmysData$distribution)


## We could reorder the data.frame following the names on the trees

## terminals <- colnames(RhinoclemmysData$distribution)

## realOrder <- match(terminals,RhinoclemmysData$tree$tip.label)

## equivalencias <- data.frame(terminals,realOrder)

## dXY2 <-  distXY

## for(cambiar in terminals){
##    dXY2$Terminal[distXY$Terminal == cambiar]     <-
##    equivalencias$realOrder[equivalencias$terminals==cambiar]
##}
```

```r
## distGraficar <- distXY

#~ distGraficar <- dXY2

## plot using ggplot

# plotDistrib <- ggplot(distGraficar,
#                       aes(x= Area, y=Terminal), size =30) +
#                 geom_point(shape=19, fill="white",
#                            color="darkgrey",
#                            size=4) +
#                 labs(title = "Distribution",
#                      y = "",
#                      x = "Area") +
#                 theme(axis.line=element_blank(),
#                       # axis.text.y=element_blank(),
#                     axis.ticks=element_blank(),
#                       # axis.title.y=element_blank(),
#                     legend.position="none",
#                     panel.background=element_blank(),
#                     panel.border=element_blank(),
#                     panel.grid.major=element_blank(),
#                     panel.grid.minor=element_blank(),
#                     plot.background=element_blank()
#                     )
#
## cowplot::plot_grid(plotTree,plotDistrib, ncol=1)

plot(plotTree)
```




```r
#~ plot(plotDistrib)

print(t(RhinoclemmysData$distribution))
```

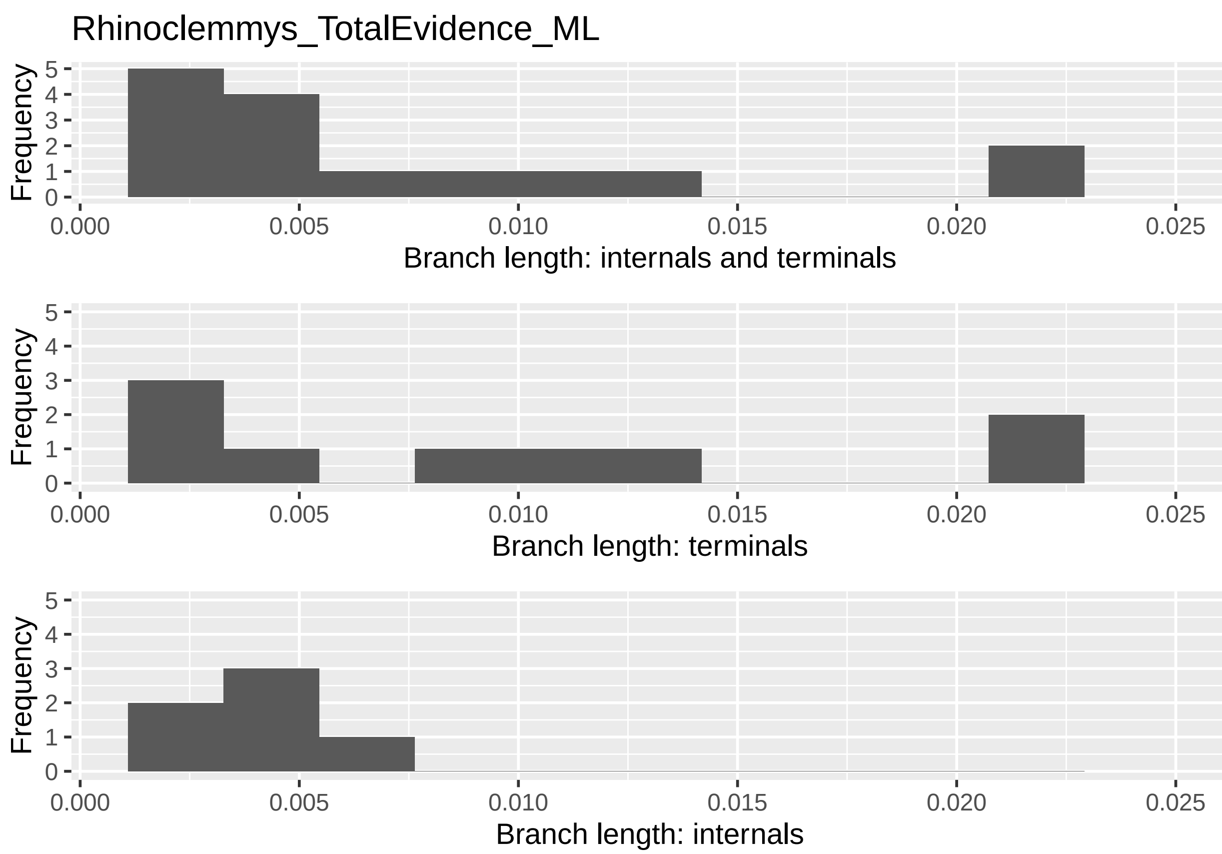
```
##                 Al An-As Cho Nsa Nuh Ph Pl
## R_annulata       1     0   1   0   0  1  0
## R_areolata       1     0   0   0   0  0  0
## R_diademata      0     0   0   1   0  0  0
## R_funerea        1     0   0   0   1  0  0
## R_melanosterna   0     0   1   0   0  1  0
## R_nasuta         0     0   1   0   0  0  0
## R_pulcherrima    0     0   0   0   1  0  1
## R_punctularia    0     1   0   0   0  0  0
## R_rubida         0     0   0   0   0  0  1
```

```r
##dev.off()
```

Now we plot the branch lengths

```r
library(ggplot2)

##par(mfrow=c(2,1))

datos <- RhinoclemmysData$tree$edge.length
data <- as.data.frame(datos)
```

```
a <- ggplot2::ggplot(data, aes(x=datos)) +
     geom_histogram(bins = 12) +
     labs(title = treeFiles, x = "Branch length: internals and terminals", y = "Frequency") +
     xlim(c(0.001, 0.025)) +
     ylim(c(0, 5))

terminals <- RhinoclemmysData$tree$edge[,2] < 1 + length(RhinoclemmysData$tree$tip.label)

datos <- RhinoclemmysData$tree$edge.length[terminals]
data <- as.data.frame(datos)

b <- ggplot(data, aes(x=datos)) +
  geom_histogram(bins = 12) +
  labs(x = "Branch length: terminals", y = "Frequency") +
  xlim(c(0.001, 0.025)) +
  ylim(c(0, 5))



datos <- RhinoclemmysData$tree$edge.length[!terminals]
data <- as.data.frame(datos)

c <- ggplot(data, aes(x=datos)) +
  geom_histogram(bins = 12) +
  labs(x = "Branch length: internals", y = "Frequency") +
  xlim(c(0.001, 0.025)) +
  ylim(c(0, 5))


  ##

cowplot::plot_grid(a, b, c, nrow=3)
```




```
##
#
#  cat("Terminals, with BL larger than 0.02 = ",
# findTerminalgivenLength(RhinoclemmysData$tree,0.02))
#
##


###dev.off()
```

The branch length histograms and the tree plot show that the internal length branches are similar among each other, and different to terminals'; there are two longer branches (larger than 0.02), *R. nasuta* (inhabiting Cho) and *R. rubida* (Pl), whereas Al and Cho are the richest.

PD values

First, I calculate the PD value for the areas, given this tree.

```
RhinoclemmysData$tablePD <- PDindex(RhinoclemmysData$tree,
                              distribution=RhinoclemmysData$distribution,
                                    root=TRUE,
                                    percentual = TRUE)

RhinoclemmysData$matrixPD <- as.data.frame(
                                  matrix(
                                      unlist(RhinoclemmysData$tablePD),
```

```
                                    nrow= length(treeFiles),
                                    byrow=TRUE))


colnames(RhinoclemmysData$matrixPD) <- row.names(RhinoclemmysData$distribution)

row.names(RhinoclemmysData$matrixPD) <- "PD value"

 new <- (t(RhinoclemmysData$matrixPD))

 ## officially I am an Idiot

values <-  new[order(new[,1])]
names  <-  rownames(new)[order(new[,1])]

print(as.data.frame(paste(names,values)))
```

```
##   paste(names, values)
## 1             Nsa 8.05
## 2           An-As 8.42
## 3             Ph 13.18
## 4            Nuh 13.86
## 5             Pl 16.81
## 6             Al 17.82
## 7            Cho 21.86
```

```
## cat("Max value is:",max(t(RhinoclemmysData$matrixPD)),"Area:",
## colnames(RhinoclemmysData$matrixPD)[which.max(t(RhinoclemmysData$matrixPD))])
```

The highest PD is for area Cho, followed by Al, the two richest areas, and the difference in PD value is not given by the richness but the species in each region.

#Effect of branch lengths in the PD

We test the effect of the branch length on the PD values by swapping terminal and/or internal branch lengths using the three available models.

```
 for( modelo in c("simpleswap","allswap","uniform") ){

        for( rama in c("terminals","internals","all") ){

            val <- swapBL(RhinoclemmysData$tree,
                          RhinoclemmysData$distribution,
                          model = modelo,
                         branch = rama
               )

         cat( "\n\t Tree=",treeFiles,"\n\t Model=",modelo,
             "\n\t Branchs swapped=",rama,"\n" )

        printBlepd(val)

         cat( "\n\n\n" )

     }
```

```
}
```

```
## model to test simpleswap reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= simpleswap
##   Branchs swapped= terminals
##
## BestInitial:Cho
##   AreaSelected Freq Percent
## 1           Al   13      13
## 2          Cho   85      85
## 3           Pl    2       2
##
##
##
## model to test simpleswap reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= simpleswap
##   Branchs swapped= internals
##
## BestInitial:Cho
##   AreaSelected Freq Percent
## 1          Cho  100     100
##
##
##
## model to test simpleswap reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= simpleswap
##   Branchs swapped= all
##
## BestInitial:Cho
##   AreaSelected Freq Percent
## 1           Al    8       8
## 2          Cho   91      91
## 3           Pl    1       1
##
##
##
## model to test allswap reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= allswap
##   Branchs swapped= terminals
##
## BestInitial:Cho
##   AreaSelected Freq Percent
## 1           Al   35      35
## 2          Cho   53      53
## 3          Nuh   10      10
## 4           Pl    2       2
```

```
##
##
##
## model to test allswap reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= allswap
##   Branchs swapped= internals
##
## BestInitial:Cho
##   AreaSelected Freq Percent
## 1          Cho  100     100
##
##
##
## model to test allswap reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= allswap
##   Branchs swapped= all
##
## BestInitial:Cho
##   AreaSelected Freq Percent
## 1           Al   37      37
## 2        An-As    4       4
## 3          Cho   33      33
## 4          Nsa    4       4
## 5          Nuh   19      19
## 6           Pl    3       3
##
##
##
## model to test uniform reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= uniform
##   Branchs swapped= terminals
##
## BestInitial:Cho
##   AreaSelected Freq Percent
## 1           Al   46      46
## 2          Cho   50      50
## 3          Nuh    4       4
##
##
##
## model to test uniform reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= uniform
##   Branchs swapped= internals
##
## BestInitial:Cho
##   AreaSelected Freq Percent
```

```
## 1          Cho  100     100
##
##
##
## model to test uniform reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= uniform
##   Branchs swapped= all
##
## BestInitial:Cho
##   AreaSelected Freq Percent
## 1            Al   38      38
## 2         An-As    1       1
## 3           Cho   53      53
## 4           Nsa    1       1
## 5           Nuh    7       7
```

The terminal branch length is critical in the decision taken, if we swap all terminal branch lengths (model= "allswap"), or if we replace them with a uniform distribution (model= "uniform"), the area selected change from Cho to Al.

As the terminal branch lengths are distributed unequally, we might suspect that the results could depend on the longest branches that inhabit Cho/Al areas.

First, we tested whether the number of replicates had any effect.

```r
for(repetir in 1:2){

            val <- swapBL( RhinoclemmysData$tree,
                           RhinoclemmysData$distribution,
                           model = "allswap",
                           branch = "terminals",
                           nTimes = 10**repetir
                         )

            printBlepd(val)

}
```

```
## model to test allswap reps 10
##
## BestInitial:Cho
##   AreaSelected Freq Percent
## 1            Al    1      10
## 2           Cho    6      60
## 3           Nuh    3      30
## model to test allswap reps 100
##
## BestInitial:Cho
##   AreaSelected Freq Percent
## 1            Al   39      39
## 2           Cho   40      40
## 3           Nsa    1       1
## 4           Nuh   19      19
## 5            Pl    1       1
```

Roughly speaking, from 1000 on the results are alike, and the largest The difference was the use of 10 or 100 replicates. As a rule of thumb, we must use at least 100 replicates, but 1000 replicates would be better.

Now, let us see if the difference in results could be assigned to the longest branches (or not).

```
## we can use a single command (*evalBranch*)

AllTerminals <- evalBranch(tree=RhinoclemmysData$tree,
                           distribution=RhinoclemmysData$distribution,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals)
```

```
##    node initialArea FinalArea Aproach  %Delta
## 1     1         Cho        Al   lower  -47.18
## 12    3         Cho        Pl   upper  116.08
## 13    4         Cho       Nsa   upper 1819.58
## 14    5         Cho     An-As   upper 1202.58
## 16    7         Cho        Al   upper  391.02
## 17    8         Cho        Al   upper   84.95
## 18    9         Cho        Pl   upper   61.32
```

```
AllInternals <- evalBranch(tree=RhinoclemmysData$tree,
                           distribution=RhinoclemmysData$distribution,
                           branchToEval="internals",
                           approach="all")

printEvalBranch(AllInternals)
```

```
##    node initialArea FinalArea Aproach  %Delta
## 13   16         Cho     An-As   upper 1529.13
```

The selected area depended on the branch length. While the internal terminals had almost no impact, if the terminal branch length of *R. nasuta* is -47.18% shorter, or if the terminal branch length is 84.95% larger for *R. areolata*, the area selected will change from Cho to Al, and if the terminal branch length of *R. rubida* is 61.32% larger, and the area selected changes from Cho to Pl.

If the tree is ultrametric, the pair of sister pairs will behave alike, and the difference will be given by the distribution.

```
## now let's see the impact of ultra-metricity

AllTerminalsU <- evalBranch(tree=force.ultrametric(RhinoclemmysData$tree),
                           distribution=RhinoclemmysData$distribution,
                           branchToEval="terminals",
                           approach="all")
```

```
## ***************************************************************
## *                          Note:                              *
## *    force.ultrametric does not include a formal method to    *
## *    ultrametricize a tree & should only be used to coerce    *
## *   a phylogeny that fails is.ultrametric due to rounding --  *
## *    not as a substitute for formal rate-smoothing methods.   *
## ***************************************************************
```

```
cat("Ultrametric tree\n")
```

```
## Ultrametric tree
```

```r
printEvalBranch(AllTerminalsU)
```

```
##    node initialArea FinalArea Aproach  %Delta
## 9     1         Cho        Al   lower  -47.97
## 11    7         Cho        Al   upper  349.96
## 12    4         Cho       Nsa   upper 1567.21
## 13    5         Cho     An-As   upper 1567.21
## 14    8         Cho        Al   upper   91.64
## 15    9         Cho        Pl   upper  132.31
## 17    3         Cho     NuhPl   upper   215.5
```

```r
printEvalBranch(AllTerminals)
```

```
##    node initialArea FinalArea Aproach  %Delta
## 1     1         Cho        Al   lower  -47.18
## 12    3         Cho        Pl   upper  116.08
## 13    4         Cho       Nsa   upper 1819.58
## 14    5         Cho     An-As   upper 1202.58
## 16    7         Cho        Al   upper  391.02
## 17    8         Cho        Al   upper   84.95
## 18    9         Cho        Pl   upper   61.32
```

```r
AllInternalsU <- evalBranch(tree=force.ultrametric(RhinoclemmysData$tree),
                           distribution=RhinoclemmysData$distribution,
                           branchToEval="internals",
                           approach="all")
```

```
## ***************************************************************
## *                          Note:                              *
## *    force.ultrametric does not include a formal method to    *
## *    ultrametricize a tree & should only be used to coerce    *
## *   a phylogeny that fails is.ultrametric due to rounding --  *
## *    not as a substitute for formal rate-smoothing methods.   *
## ***************************************************************
```

```r
printEvalBranch(AllInternalsU)
```

```
##      node initialArea FinalArea Aproach %Delta
## NA   <NA>        <NA>      <NA>    <NA>   <NA>
## 8      16         Cho  An-AsNsa   upper 1981.8
## NA.1 <NA>        <NA>      <NA>    <NA>   <NA>
```

```r
printEvalBranch(AllInternals)
```

```
##    node initialArea FinalArea Aproach  %Delta
## 13   16         Cho     An-As   upper 1529.13
```

Literature cited

Le, Minh, and William P. McCord. 2008. "Phylogenetic relationships and biogeographical history of the genus Rhinoclemmys Fitzinger, 1835 and the monophyly of the turtle family Geoemydidae (Testudines: Testudinoidea)." *Zoological Journal of the Linnean Society* 153 (4): 751–67. https://doi.org/10.1111/j.1096-3642.2008.00413.x.

Romero-Alarcon, L. Viviana. 2020. "A total-evidence phylogeny of the crown and stem-groups of turtles (Pantestudines: Testudinata)." Master's thesis, Colombia: Escuela de Biología, UIS.